



Examiners' Report Summer 2026

Pearson Edexcel GCE

In Further Mathematics (9FM0)

Paper 3D Decision Mathematics 1

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General comments

The paper was accessible to learners, with many demonstrating sound knowledge across a range of topics. Many responses were well presented and made effective use of the diagrams and graphs printed in the answer booklet.

Learners should be reminded of the importance of setting out their methods clearly, as Decision Mathematics is a method-based examination. Some responses did not use the method required by the question, or did not include accurate diagrams and graphs, which limited the number of marks available.

Comments on individual questions

Question 1

Part (a)

Part (a) was answered well. The learners chose either Middle Right or Middle Left pivots to approach this question. The responses which sorted into descending order were not gaining the final two marks. There were some instances of miscopying or missing a number, which invariably also led to two marks not being awarded; in most cases this was when the passes were not laid out clearly. Some errors were seen with inconsistent pivot choice, usually on the third pass, although those responses that choose middle left had the advantage that they got all central pivots if they did each pass correctly. There were also errors such as moving an item too early (some responses moved 37 to the far left after their first pass). There were also some responses that were not aware of the difference between quick and bubble sorts and some tried variations on shuttle sorts. Some responses used arrows to show the position of an item once it had been fixed; these responses must ensure that they write out their sorted list in full at the end.

Part (b)

Part (b) was also generally answered well, with most learners writing out their full list of arcs and writing accept/reject next to each one. Some responses did not make the acceptance/rejection choice (or their order of choosing arcs) clear, which resulted in marks not being awarded in this part. Some responses were not awarded marks for writing a full ordered list and then the MST, but without making their rejections explicit. Drawing a diagram is not being required in this part; most learners that did this avoided the errors of adding too many arcs or creating a cycle. There were a small number of learners who seemed to try to show the rejections on their sorted values from part (a), but omitted to make their method clear (and was not in part (b)), and so were not awarded any marks. The final mark was earned by the most of the learners as, even those who did not demonstrate a complete application of Kruskal's algorithm, but still obtained a correct MST.

Question 2

Part (a)

This route inspection problem was accessible to many learners and provided effective differentiation at the lower end of the grade spectrum. A small number of responses did not identify the problem type or the required technique of pairing the odd nodes, so no marks were awarded for this question.

Most learners identified the required algorithm and provided three pairings of the correct four odd nodes, with at least one total, usually the optimal one. To gain full credit, responses needed to show the full algorithmic process: finding all three totals before selecting the optimal, minimum value. Responses that only calculated '600' did not demonstrate a choice between the three possible totals, so the marks for finding the correct totals and for selecting the optimal value could not all be awarded.

A good proportion of learners found at least two correct totals and were able to give the correct total length of inspection route, along with the correct arcs to be traversed twice. Learners listing paths rather than arcs to repeat usually progressed well on this question.

Part (b)

Part (b) was a standard amendment to the route inspection problem. Responses needed to recognise that, when a route starts and finishes at two different nodes, only the shortest path connecting the other two odd nodes needs to be repeated.

Many learners selected the shortest of the six pairings found in part (a), usually EH, and recognised that this meant the start and finish nodes were B and F, respectively. A small number of responses interpreted EH as the start and finish nodes because it was the path to be repeated, so the final accuracy mark could not be awarded.

Question 3

Part (a)

Part (a)(i) proved to be more challenging. Many learners calculated a lower bound for the number of bins by dividing the sum of the values by the bin size, but did not always provide a complete conclusion.

Some responses included detailed explanations of how the values could be combined, but these were sometimes only partially correct because they did not include the critical wording. Responses that recognised there were five numbers greater than or equal to half the bin size did not always state both “greater than or equal to” and “half the bin size”, both of which were needed for a fully correct explanation.

Responses to part (a)(ii) generally showed familiarity with the first fit bin-packing algorithm, with the first five values usually packed correctly. The remaining values were also often packed correctly, although some responses placed ‘10’ in bin 4 rather than in the correct bin.

Part (b)

The time estimate in part (b), based on the algorithm having quadratic order, was usually calculated correctly. In a small number of responses, the algorithm was treated as having linear or quartic order.

Question 4

Part (a)

Part (a) was answered very well, with only a small number of responses not gaining the mark because of a slip or leaving a cell blank.

Part (b)

Part (b) was answered well, with most errors appearing to be slips rather than misconceptions. Where the method mark was not awarded, this was usually because one or more cells had been left blank. It was uncommon for this mark not to be awarded because of changes made in the 3rd/4th row/column.

If one cell was left blank in an iteration, the method and accuracy marks for that iteration could not both be awarded, even when the other 15 cells were correct. Accuracy errors, where they occurred, were generally similar in nature. For example, some responses gave a value of 17 rather than 27 for the distance from A to E after iteration 3, or 17 for the distance from E to A after iteration 4. This appeared to result from using the highlighted 7 for DA together with the incorrect value of 10 for EC.

Part (c)

Part (c) scored well. The most common errors were not returning to A at the end of the route and starting with AB rather than AD. Among those who had made slips in (b) and were working from an incorrect final table, some missed out by having their "from" and "to" reversed, generally by ending with their weight for AE which differed from their EA. It is likely that some learners who scored full marks were also running NN with rows and columns transposed, but the fact that the only cells in the correct final matrix not symmetrical over the main diagonal (AB/BA) don't feature in the correct solution meant that this was not detectable for those responses who didn't show working on a table. Learners should be reminded that, when using the final Floyd table for the nearest neighbour algorithm, they should read along the rows rather than down the columns.

A small number of responses did not gain the final mark because the nearest neighbour route was doubled after a correct answer had already been found. These responses appeared to confuse the nearest neighbour method with the use of a minimum spanning tree to obtain an upper bound.

Question 5

Part (a)

Despite the algebraic nature of the question, most learners attempted it and performed well. Some responses found it challenging to show clearly that the working values had been entered in the correct order, partly because of the size of the boxes provided for these values. This was taken into account during marking. Nearly all learners gained the method mark, and many responses gained the first four marks. Where marks were not awarded, this was often because working values were missing.

The route proved challenging in some responses, with nodes being omitted – a few just missed one node but others seemed to be unable to follow the route appropriately. Nearly all were able to follow through their final value at K to gain the last accuracy mark.

Part (b)

Nearly all learners were able to access part (b), with many applying the correct method. Where the answer marks were not awarded, this was often because incorrect values from part (a) had been used.

A small number of responses did not give a solution for x from their equation, so the method mark could not be awarded. Once the equation had been solved, most responses selected the correct value of x . Some responses did not go on to calculate the time taken to travel to K, so the final mark could not be awarded.

Question 6

Part (a)

In part (a), almost all learners gained the first two marks. The few errors seen were usually sign errors involving the surplus and artificial variables. There were a few more errors in finding I, with some responses having the signs reversed, some making algebraic errors in substituting for a_1 and a_2 , and a few attempting a big M method. A few responses also did not rearrange P. These errors were always reproduced in the tableau, and a few more responses with correct I and P missed the last mark by giving incorrect basic variables in the tableau (generally s_1, s_2, s_3).

Part (b)

Responses to part (b) varied. Many gave a correct answer but also included additional observations that were not relevant to the question. Some responses stated that there were “no negative coefficients on the I row”, or equivalent, which was not sufficient to gain the mark.

Part (c)

Part (c) was generally answered well. A small number of responses used an incorrect pivot, so the remaining marks for the iteration could not be awarded.

Most responses that selected the correct pivot also changed the basic variable, gaining the B mark, and went on to gain at least the method mark. Where the accuracy mark was not awarded, this was usually due to arithmetic errors.

Some responses omitted the row operations or used incorrect row operations. A common omission was not specifying which row or expression was being multiplied by $\frac{7}{3}$ to obtain the new first row.

Part (d)

For responses that could still access the final two marks, the majority gained them. A few responses gave “positive” rather than “non-negative”, did not correctly identify the objective row, omitted a final value, or could not use the follow-through because one of their values was negative.

A small number of responses referred to a profit column rather than a profit row. There were very few references to “negative variables”, despite this being mentioned in the notes section of the mark scheme.

Some responses recalculated P from the profit equation rather than reading it directly from the tableau, which created additional work. Some responses did not state the value of P.

Question 7

This non-standard Critical Path Analysis problem provided effective differentiation, with a wide range of marks seen. A number of responses fully solved the problem and gained full marks throughout.

Part (a)

Part (a) asked learners to draw an activity network. Whilst the usual range of errors (forgetting arrows on activities, missing an activity, adding additional dummies) were all seen, many responses were able to draw a fully correct network, with the most typical way not to gain a mark being the addition of an unnecessary dummy (usually after K or L).

Part (b)

Part (b) provided the most differentiation in this question through a variation on a typical lower bound calculation. Many responses calculated 75/25 rather than

recognising that some of the times needed to be doubled, so neither mark could be awarded.

A reasonable proportion of responses recognised the need to double the relevant times and gained both marks. The award of one mark was uncommon and was generally seen where a numerical slip had been made.

Part (c)

Part (c) asked learners to construct a resource histogram. Although this is a different tool than Gantt charts and scheduling diagrams, many responses showed a clear understanding of the requirements and included an accurate diagram, gaining full marks.

There were very few instances of overhangs, where the method mark could not be awarded. More commonly, responses either did not extend to time 20 or histograms showed too few workers, most often between times 7 and 12, which meant the method mark could not be awarded.

Many responses gained all the marks. The most frequent error was showing 3 workers rather than 4 workers between times 17 and 20. Overall, this part was answered well.

Question 8

This multi-part linear programming question allowed learners to gain marks in some parts independently of errors or omissions in others.

Part (a)

In part (a), the first four marks were accessible to most learners. Some responses omitted the word “maximise” from the objective function, while others left the area inequality with non-integer coefficients. A small number of responses made slips in the final two constraints, either by using an incorrect variable or by reversing the inequality.

Part (b)

Some learners did not attempt part (b). Many responses calculated “c” correctly by recognising that 800 was the sum of the unknowns, before obtaining the expression for z and substituting it into the expression for P found in part (a).

Other successful responses partitioned the expression for P in order to substitute 800 directly. A small number of responses did not include $P =$ in their working, which was required for this “show that” response.

Part (c)

Many responses to part (c) included clear, accurate and fully correct graphs, with the three lines from the inequalities on the question paper plotted, appropriate shading shown, and “R” labelled. Light shading, or shading limited to areas adjacent to the boundary lines, generally made responses easier to mark. Heavy shading across excluded regions could make it more difficult for examiners to see the plotted lines and feasible region clearly. Learners should be reminded that lines need to be **plotted accurately**. Inaccurate plotting not only affected the marks available for the graph itself, but could also affect later marks that depended on identifying the correct feasible region.

Part (d)

Part (d) (i) & (ii) – the question says “Use the objective line method”. For this method it is essential that an accurate objective line be drawn on the graph. Without such a line the final seven marks cannot be awarded, as **the line is the basis of the method**. Also no marks can be awarded for responses using the vertex testing method. Those responses who drew the reciprocal gradient objective line were eligible for just two of the final seven marks. Following on from a correct objective line many responses were seen with the correct optimum point identified, usually calculated and sometimes shown on their graph. Many learners did not make progress from this point.

The final four marks depended on integer point testing around the correct optimum point. Many responses tested the four points closest to the optimum, leading to (116, 617) as the best point considered. However, (117, 615) was also within the feasible region and produced the maximum profit.

Very few responses gave this correct solution, as it required problem-solving beyond the routine approach. Many responses did not explicitly show point testing in the two relevant inequalities, so the method mark could not be awarded.

Learners should be encouraged to list the points and constraints in a table to support accurate point testing. Responses that clearly demonstrated point testing were able to gain the final two marks through a correct follow-through answer in context, together with a profit calculation.